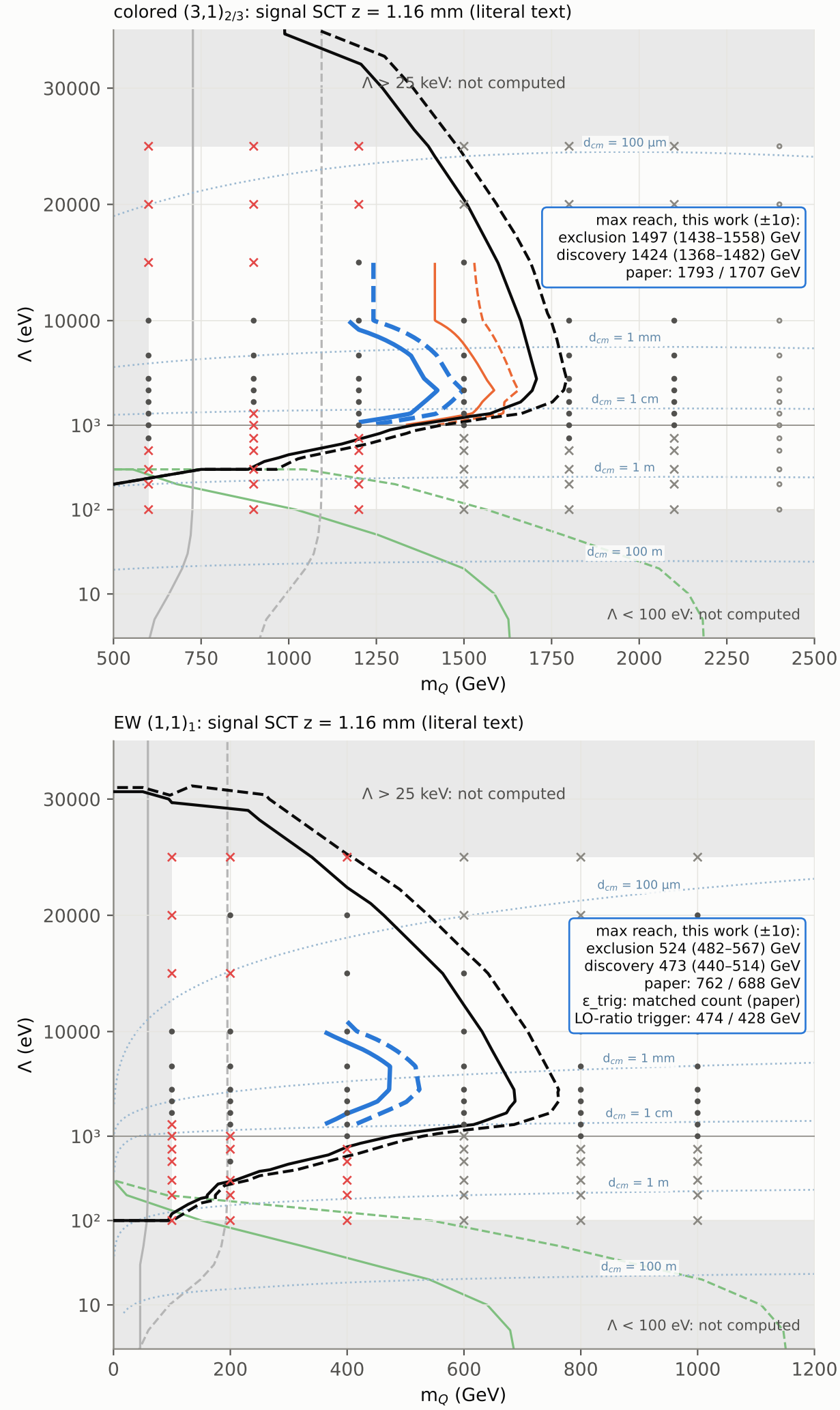
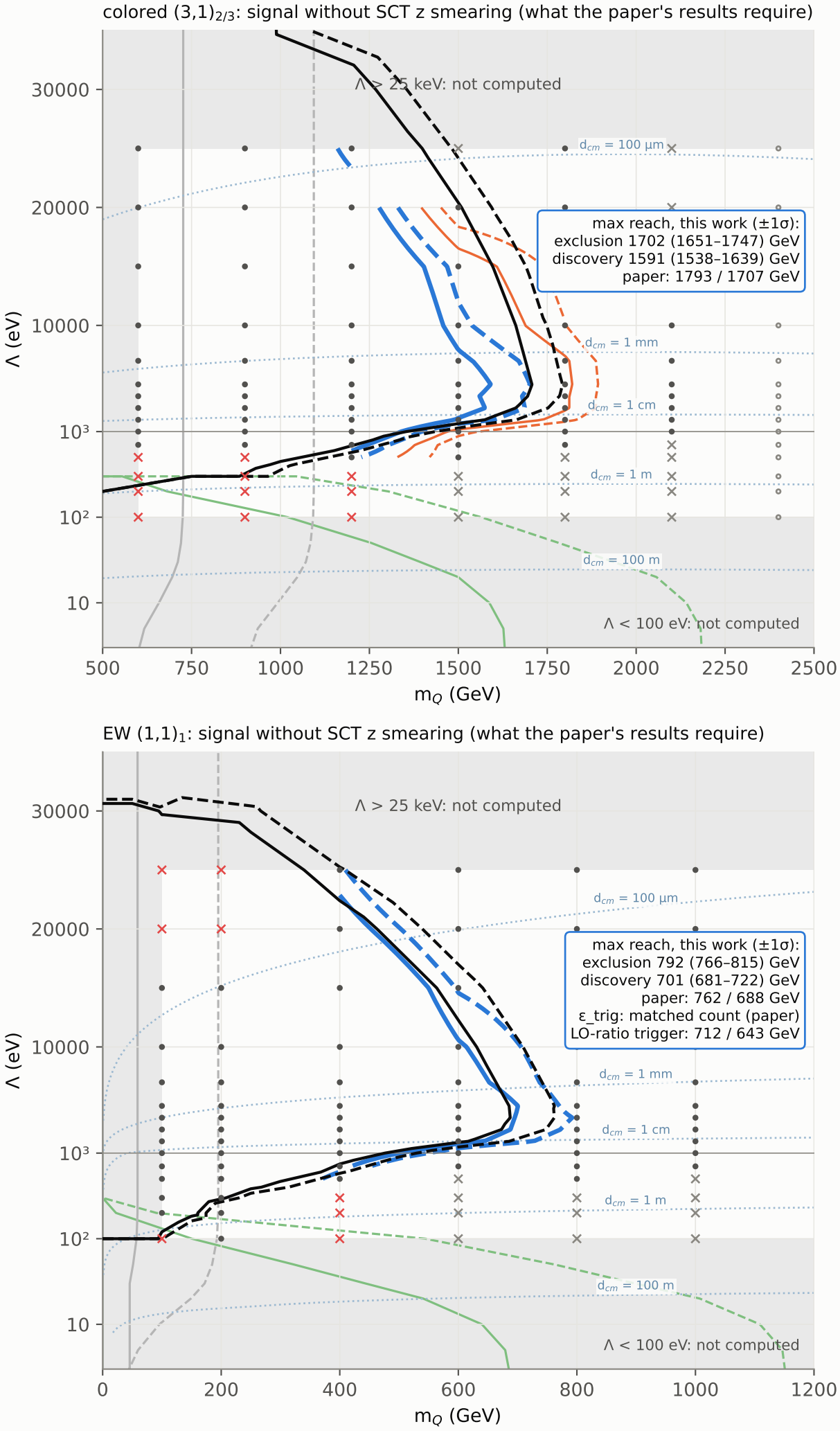
Figure 6 of arXiv:1708.02243: contours of 3 (exclusion) and 5 (discovery) signal events at 300 fb⁻¹, no background, as in the paper
left: signal SCT z = 1.16 mm (literal text); right: signal without SCT z smearing (what the paper's results require)





paper: exclusion (3 events)
paper: discovery (5 events)
this work: exclusion (3 events)
this work: discovery (5 events)
this work, colored, no charge factor: exclusion
this work, colored, no charge factor: discovery
HSCP 12.9 / 300 fb⁻¹ (from the paper)
monojet 12.9 / 300 fb⁻¹ (from the paper)
d_{cm} (from the paper)
grid point with ≥ 1 event in the signal region
0 events passed: N < 3 at 95% CL
0 events passed: not resolved (95% UL on N ≥ 3)
not propagated: N < 3 even with $\varepsilon_{fid} = \varepsilon_{reco} = 1$

Normalisation: σ = MadGraph5 3.7.3 LO (0 + 1 jet, MLM-matched with Pythia8) for one quirk species, no K-factor; the paper gives no σ. The "N_Q = 2" of the paper's panel titles is NOT applied (no factor 2 on σ): it appears only in the titles of the paper's Fig. 6 panels (“N_Q = 2 Quirks”, arXiv v1 and v2) and is never defined or used in the text, so it is not known whether it means a factor on σ (e.g. two quirk species) or only the two quirks of each pair; σ × 2 is shown as a sensitivity in F6_kfactor. Colored: a factor 1.5–1.7 on σ brings our maxima close to the paper's; EW: our maxima are already slightly above the paper's (+2–4%; K ≈ 0.85–0.9 would match), so one common K-factor cannot explain both channels (sensitivity, not a fit; F6_kfactor).
EW TRIGGER CHANGED (2026-10-19): ε_trig is now the matched + shower count N(p_T($Q\bar{Q}$) > 200 GeV) / N(matched) of the same sample that gives ε_fid and ε_reco (152/1024 = 0.148 at 800 GeV in this Fig. 6 sample; 186/1045 = 0.178 in the independent Table I sample, same generator settings, 1.8σ apart; combined 338/2069 = 0.163, vs the paper's 0.10), the method the paper describes. Before, it was the LO ratio σ($Q\bar{Q}$ + j, p_T(j) > 200) / σ($Q\bar{Q}$) (0.096 at 800 GeV, = Table I's 0.10; not stated in the paper), 0.54–0.75 × the matched count. This moves the EW maximum outward from 712 / 643 to 792 / 701 GeV (exclusion / discovery, no SCT z smearing; +80 / +58 GeV, as the earlier note's estimate of ~60–80 GeV). Colored: ε_trig = matched count (unchanged); the both-charged factor (~0.33) is included (thin orange: without it).
Red ×: at low Λ for colored 600–1200 GeV and EW 100–200 GeV our samples (40–70 events per mass) are too small; the paper's lower-left edge passes there and is not tested here (larger samples needed). Λ = 15–25 keV: computed (the integrator's 20-million-step ceiling removes ≤ 13% of the events fiducial at 10 keV there); Λ > 25 keV: not computed, because that ceiling removes 5–56% of them at 30–35 keV (needs a larger ceiling and more memory). The maxima are at 3–5 keV and are unaffected.
Annihilation is not simulated (as in the paper). Worst case (α = 0.05, annihilation chance at every second meeting) it lowers the signal near the maxima by ~17–24% (EW 4–5 keV) and ~17% (colored 5 keV), moving the maxima inward by ~30 GeV (colored) and ~25–40 GeV (EW).